

Dr Waseem Raza, PhD

Machine Learning Engineer | Financial AI/ML & Data Science

Financial ML | Forecasting | Risk Analytics | Data Engineering

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Professional Summary

Machine Learning Engineer and Data Scientist with a PhD in Electrical and Computer Engineering, production ML experience, and finance-focused portfolio work in credit-risk modeling, loan-default prediction, time-series forecasting, and reinforcement-learning study. Brings strong foundations in statistical learning, model evaluation, cloud data engineering, and reproducible analytics while developing deeper specialization in machine learning for financial prediction, portfolio management, trading, and risk.

Technical Expertise, Leadership, and Professional Strengths

Financial Machine Learning: Credit-risk modeling, loan-default prediction, supervised learning, predictive modeling, reinforcement-learning concepts, portfolio and trading applications

Time Series & Statistics: Forecasting, statistical modeling, feature engineering, regression, classification, model diagnostics, validation, and uncertainty-aware analysis

Data Science & Analytics: Python, SQL, Pandas, NumPy, SciPy, Statsmodels, exploratory analysis, visualization, and data-quality assessment

Machine Learning: scikit-learn, XGBoost, TensorFlow, PyTorch, cross-validation, hyperparameter tuning, anomaly detection, and model evaluation

Data Engineering & Cloud: ETL/ELT, Azure Databricks, Apache Spark, PySpark, Azure Data Factory, Synapse Analytics, BigQuery, and AI-ready datasets

MLOps & Engineering: Git, Docker, CI/CD, APIs, testing, reproducible workflows, model deployment, and production-oriented engineering

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

Feb. 2025-Present

Bedminster, NJ

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Designed domain-adaptive neural models and reusable data-processing workflows that generalize across heterogeneous operational domains and reduce performance degradation caused by distribution shifts.
- Developed multimodal AI prototypes combining work-order text, location information, network context, and operational signals to support GenAI-assisted diagnostics for VoNR and 5G service assurance.
- Created model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable deployment in high-impact operational environments.
- Partnered with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and production-oriented data assets.

Data Scientist and AI/ML Engineer

Impac Exploration Services

Sep. 2024-Dec. 2024

Weatherford, OK

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.
- Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.

Graduate Research Assistant

AI4Networks, University of Oklahoma

Sep. 2019-Sep. 2024

Tulsa, OK

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.

- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering	09/2019-07/2025
University of Oklahoma, Norman, Oklahoma, USA	
MSc, Telecommunication Engineering	08/2014-08/2016
University of Engineering and Technology, Taxila, Pakistan	
BSc, Telecommunication Engineering	11/2010-06/2014
University of Engineering and Technology, Taxila, Pakistan	

Selected Projects

Financial ML and Forecasting	2024-2025
Portfolio Project	
• Credit-risk and loan-default modeling alongside time-series forecasting workflows for financial data.	
Tools & methods: Python, Machine Learning, Time Series, Model Evaluation	
AI-Ready Lakehouse Pipeline	2024
Portfolio Project	
• An Azure Databricks data pipeline designed for analytics and downstream machine-learning workloads.	
Tools & methods: Azure, Databricks, Spark, Data Engineering	
System Design Case-Study Portfolio	2026 / In progress
Guided Project	
• Documented architecture studies that translate product requirements into scalable services, data models, APIs, caching strategies, and reliability tradeoffs.	
Tools & methods: System Design, APIs, Distributed Systems, Scalability	

Certifications and Continuing Education

1. Machine Learning and Reinforcement Learning in Finance Specialization — New York University / Coursera Completed (0/4) courses	
Focus: An applied machine-learning specialization focused on financial prediction, portfolio management, trading, risk modeling, market dynamics, and reinforcement-learning methods for finance.	
Tools & Skills: Machine Learning for Finance, Reinforcement Learning, Financial Modeling, Portfolio Management, Financial Trading, Risk Modeling, Predictive Modeling, Supervised Learning, Unsupervised Learning, Statistical Machine Learning, Market Dynamics, Derivatives, TensorFlow, Scikit-learn, Python	
2. Machine Learning — Stanford University / Coursera Completed	
Focus: Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.	
Tools & Skills: Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization	
3. Google Cloud Data Analytics Professional Certificate — Google Cloud / Coursera Completed	
Focus: Cloud data transformation, analysis, visualization, and decision support using Google Cloud tools.	
Tools & Skills: SQL, BigQuery, Data Analytics, Visualization	